

VIJAYA VITTALA INSTITUTE OF TECHNOLOGY

Department of Mathematics

QUESTION BANK

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Prepared By

Mr.Ramesh

Department of Mathematics

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Department of Mathematics

MODULE - 1

INTRODUCTION TO NUMERICAL METHODS

Errors and their Computations

1. The value $e = 2.71828$ and approximate value is 2.72. Find relative error, absolute error, and percentage error.
2. Define the following:

(a) Round off Error	(b) Inherent Error	(c) Truncation Error
(d) Absolute Error	(e) Relative Error	(f) Percentage Error
3. Obtain a second degree polynomial approximation to

$$f(x) = (1 + x)^{1/2}, \quad x \in [0, 0.1]$$

using Taylor series about $x = 0$. Approximate $f(0.05)$ and find a bound of the truncation error.

4. The true value $\pi = 3.141592$ and approximate value is 3.14. Find relative error, absolute error, and percentage error.
5. Find the absolute error if the number 0.00545828 is
 - (a) truncated to 3 decimal digits,
 - (b) rounded off to 3 decimal digits.
6. Round off 865240 and 37.463 to four significant figures and calculate absolute and percentage error.
7. If

$$u = \frac{4x^2y^3}{z^4},$$

and the error in x, y, z is 0.001, compute the error in u when $x = y = z = 1$.

8. If

$$R = 10x^3y^2z^2,$$

and relative errors in x, y, z are 0.03, 0.01, 0.02 respectively at $x = 3, y = 1, z = 2$, find absolute, percentage, and relative error in R .

Bisection Method

1. Find the root of

$$x^3 - 4x - 9 = 0$$

using the Bisection method correct to three decimal places.

2. Find the real root of

$$f(x) = x^3 - 5x + 1 = 0$$

correct to three decimal places.

3. Find the real root of

$$f(x) = \cos x - xe^x = 0$$

correct to three decimal places.

4. Find the real root of

$$f(x) = x^3 - 2x - 1 = 0$$

using the Bisection method. Perform five iterations.

Regula Falsi Method

1. A real root of

$$f(x) = x^3 - 5x + 1 = 0$$

lies in $(0, 1)$. Perform four iterations of the Regula-Falsi method to obtain this root.

2. Find the real root of

$$f(x) = x^3 - 2x - 5 = 0$$

correct to three decimal places using Regula Falsi method.

3. Compute the real root of

$$x \log x - 1.2 = 0$$

correct to three decimal places using Regula Falsi method.

4. Find the real root of

$$\cos x = 3x - 1$$

correct to three decimal places using Regula Falsi method.

Newton-Raphson Method

1. Find the real root of

$$3x = \cos x + 1$$

by the Newton-Raphson method.

2. Find a real root of

$$x^3 + x^2 + 3x + 1 = 0$$

by performing two iterations using Newton-Raphson method.

3. Find the cube root of 37 correct to three decimal places using Newton-Raphson method.

4. Find the real root of

$$xe^x - 2 = 0$$

correct to three decimal places using Newton-Raphson method.

Secant Method

1. Find the real root of

$$x^3 - 2x - 5 = 0$$

correct to three decimal places using Secant method.

2. Obtain the real root of

$$xe^x = 2$$

correct to four decimal places using Secant method.

MODULE - 2

NUMERICAL SOLUTIONS FOR SYSTEM OF LINEAR EQUATIONS

Vector Norms

1. If

$$x = (3, -4, 5),$$

find the L_1 , L_2 , and L_∞ norms of x .

2. If

$$x = [-1, 4, 2],$$

calculate the L_1 , L_2 , and L_∞ norms of x .

3. If

$$x = (2, -1, 3, -4),$$

find the L_1 , L_2 , and L_∞ norms of x .

Matrix Norms

1. Find the L_1 , L_2 , and L_∞ norms of

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}.$$

2. Find the matrix norms of

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}.$$

3. Calculate the matrix norms of

$$A = \begin{bmatrix} 7 & 1 & 8 \\ 4 & 5 & 8 \\ 10 & 4 & 2 \end{bmatrix}.$$

Condition Number

1. Find the condition number of

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$$

with respect to the L_1 norm.

2. Find the condition number of

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}.$$

3. Find the condition number of

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}.$$

4. Find the condition number of

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}.$$

5. Find the condition number of

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

6. Find the condition number of

$$A = \begin{bmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{bmatrix}.$$

Ill-Conditioned System

1. Show that the following system is ill-conditioned:

$$2x + y = 4, \quad 2x + 1.01y = 4.$$

Gauss–Seidel Method

1. Solve the consistent system

$$x_1 + 3x_2 = 4, \quad 2x_1 + 5x_2 = 7.$$

2. Solve the consistent system

$$x + y + z = 6, \quad x - y + z = 2, \quad 2x + y - z = 1.$$

3. Solve the consistent system

$$x + 2y + 3z = 14, \quad 3x + y + 2z = 11, \quad 2x + 3y + z = 11.$$

4. Using Gauss–Seidel method, solve

$$10x + 2y + z = 9, \quad 2x + 20y - 2z = -44, \quad -2x + 3y + 10z = 22,$$

starting from $(0, 0, 0)$ and perform three iterations.

5. Solve the following system by Gauss-Seidel method:

$$10x + y + z = 12, \quad x + 10y + z = 12, \quad x + y + 10z = 12.$$

6. Solve the following system by Gauss-Seidel method correct to three decimal places:

$$\begin{aligned} x + y + 54z &= 110, \\ 27x + 6y - z &= 85, \quad 6x + 15y + 2z = 72. \end{aligned}$$

LU Decomposition

1. Solve by LU decomposition (Cholesky method):

$$\begin{aligned} 3x + 2y + 7z &= 4, \\ 2x + 3y + z &= 5, \quad 3x + 4y + z = 7. \end{aligned}$$

2. Solve the following system using LU decomposition:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 9, \\ 4x_1 + 5x_2 + 6x_3 &= 24, \\ 3x_1 + x_2 - 2x_3 &= 4. \end{aligned}$$

3. Solve the following system using LU decomposition:

$$\begin{aligned} x + 3y + 8z &= 4, \\ x + 4y + 3z &= -2, \quad x + 3y + 4z = 1. \end{aligned}$$

Rayleigh Power Method

1. Find the dominant eigenvalue and corresponding eigenvector of

$$A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$$

using the Rayleigh power method. Take

$$X^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and perform four iterations.

2. Find the dominant eigenvalue and eigenvector of

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

using the power method.

3. Find the dominant eigenvalue and corresponding eigenvector of

$$A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$$

using the power method. Take

$$X^{(0)} = \begin{bmatrix} 1 \\ 0.8 \\ -0.8 \end{bmatrix}$$

and perform five iterations.

Jacobi Method

1. Find the eigenvalues of

$$A = \begin{bmatrix} 7 & 1 \\ 1 & 7 \end{bmatrix}$$

using Jacobi method.

2. Find the eigenvalues of the symmetric matrix

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$$

using Jacobi method.

3. Find the eigenvalues of

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

using Jacobi method.

MODULE - 3

INTERPOLATION

1. Using Newton's forward interpolation formula, find the value of $f(2.5)$ from the following table:

x	1	2	3	4	5
$f(x)$	1	8	27	64	125

2. Using Newton's forward interpolation formula, find $f(1.5)$ from the following table:

x	0	1	2	3	4
$f(x)$	1	2	9	28	65

3. Find the value of $f(15)$ using Newton's forward interpolation formula from the following table:

x	10	12	14	16	18
$f(x)$	100	144	196	256	324

Newton's Backward Difference Formula

1. Using Newton's backward interpolation formula, find $f(3.5)$ from the following table:

x	1	2	3	4	5
$f(x)$	1	8	27	64	125

2. Using Newton's backward interpolation formula, find $f(3.5)$ from the following table:

x	1	2	3	4	5
$f(x)$	2	5	10	17	26

3. Estimate the value of $f(24)$ using Newton's backward interpolation formula:

x	20	21	22	23	24
$f(x)$	0.3420	0.3584	0.3746	0.3907	0.4067

Newton's Divided Difference Formula

1. Using Newton's divided difference interpolation formula, find the polynomial passing through the points

$$(1, 1), \quad (2, 4), \quad (3, 9), \quad (4, 16).$$

2. Find the value of $f(2.5)$ using Newton's divided difference formula from

x	1	2	3	4
$f(x)$	1	8	27	64

3. Construct the interpolating polynomial using Newton's divided difference formula for the points

$$(0, 1), \quad (1, 3), \quad (2, 7), \quad (4, 21).$$

Lagrange's Interpolation Formula

1. Using Lagrange's interpolation formula, find the value of $f(2.5)$ from

x	1	2	3	4
$f(x)$	1	8	27	64

2. Find the polynomial passing through the points

$$(1, 1), \quad (2, 4), \quad (3, 9)$$

using Lagrange's interpolation formula.

3. Using Lagrange's interpolation formula, find $f(2)$ from

x	0	1	3	4
$f(x)$	1	3	19	35

Piecewise Linear and Quadratic Interpolation

1. Use piecewise linear interpolation to estimate the value of $f(2.5)$ from the given data:

x	1	2	3	4	5
$f(x)$	1	4	9	16	25

2. Use piecewise quadratic interpolation to estimate the value of $f(2.5)$ from the given data:

x	1	2	3	4	5
$f(x)$	1	4	9	16	25

MODULE - 4

DIFFERENTIAL EQUATIONS OF FIRST AND HIGHER ORDER

Bernoulli's Differential Equations

1. Solve the Bernoulli's differential equation

$$\frac{dy}{dx} + \frac{y \log y}{x} = y \left(\frac{\log y}{x} \right)^2.$$

2. Solve

$$\frac{dz}{dx} + \left(\frac{z}{x} \right) \log z = \frac{z}{x} (\log z)^2.$$

3. Solve

$$\frac{dy}{dx} + \frac{y}{x} = x^2 y^6.$$

4. Solve

$$\frac{dy}{dx} - y \tan x = \frac{\sin x \cos^2 x}{y^2}.$$

5. Solve

$$(x^3 y^2 + x y^2) dx = dy.$$

Homogeneous Differential Equation

1. Solve the homogeneous differential equation

$$\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = 0.$$

2. Solve

$$\frac{d^4 y}{dx^4} + 8 \frac{d^2 y}{dx^2} + 16y = 0.$$

3. Solve

$$(D^3 + 1)y = 0.$$

4. Solve

$$4 \frac{d^4 y}{dx^4} - 8 \frac{d^3 y}{dx^3} - 7 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = 0.$$

5. Solve

$$(D^4 + 4D^3 - 5D^2 - 36D - 36)y = 0.$$

Reducible to Exact Differential Equation

1. Solve

$$y(x + y + 1) dx + x(x + 3y + 2) dy = 0.$$

2. Solve

$$(x^2 + y^3 + 6x) dx + y^2 x dy = 0.$$

3. Solve

$$y(2xy + 1) dx - x dy = 0.$$

4. Solve

$$(4xy + 3y^2 - x) dx + x(x + 2y) dy = 0.$$

5. Solve

$$(x^2 + y^2 + x) dx + xy dy = 0.$$

6. Solve

$$(y^4 + 2y) dx + (2y^3 + 2y^4 - 4x) dy = 0.$$

Non-Homogeneous Differential Equation

1. Solve

$$6 \frac{d^2 y}{dx^2} + 17 \frac{dy}{dx} + 12y = e^{-x}.$$

2. Solve

$$y'' + 2y' + y = \cosh\left(\frac{x}{2}\right).$$

3. Solve

$$\frac{d^3 y}{dx^3} + y = \cos(2x + 1) + \dots$$

MODULE - 5

NUMERICAL INTEGRATION AND NUMERICAL SOLUTION OF DIFFERENTIAL EQUATIONS

Trapezoidal Rule, Simpson's Rules and Weddle's Rule

1. Evaluate

$$\int_4^{5.2} \log x \, dx$$

using Trapezoidal, Simpson's $\frac{1}{3}$, Simpson's $\frac{3}{8}$, and Weddle's rules with 7 ordinates (or 6 strips).

2. Evaluate

$$\int_0^6 x e^{-0.5x} \, dx$$

using the Trapezoidal rule with 3 strips, correct to 3 decimal places.

3. Evaluate

$$\int_0^1 \frac{x \, dx}{1+x^2}$$

taking seven ordinates by Simpson's $\frac{1}{3}$ rule and Weddle's rule.

4. Evaluate

$$\int_0^{\pi/2} \sqrt{\cos \theta} \, d\theta$$

by dividing the interval into 6 equal parts using Simpson's $\frac{3}{8}$ and Weddle's rules.

5. Evaluate

$$\int_0^6 3x^2 \, dx$$

by dividing $[0, 6]$ into six equal parts using Simpson's $\frac{1}{3}$, Simpson's $\frac{3}{8}$, and Trapezoidal rules.

Taylor's Series

1. Use Taylor's series method to find $y(0.2)$ from

$$\frac{dy}{dx} = 2y + 3e^x, \quad y(0) = 0.$$

2. Find y at $x = 0.1$ and $x = 0.2$, correct to five decimal places, using Taylor's series method:

$$\frac{dy}{dx} = x^2 y - 1, \quad y(0) = 1.$$

3. Use Taylor's series method to solve

$$\frac{dy}{dx} = \log(xy),$$

given $y(1) = 2$, and find $y(1.1)$ and $y(1.2)$.

-
4. Use Taylor's series method to solve

$$\frac{dy}{dx} = x - y,$$

given $y(0) = 1$, and find $y(0.1)$ up to the fourth-degree term.

Runge–Kutta 4th Order

1. Use the fourth-order Runge–Kutta method to solve

$$(x + y)\frac{dy}{dx} = 1, \quad y(0.4) = 1,$$

and find $y(0.5)$ correct to 4 decimal places.

2. Use the fourth-order Runge–Kutta method to solve

$$\frac{dy}{dx} = 3x + \frac{y}{2}, \quad y(0) = 1,$$

taking $h = 0.2$.

3. Apply the fourth-order Runge–Kutta method to find y at $x = 0.2$, given

$$\frac{dy}{dx} = x + y, \quad y(0) = 1.$$

4. Use the Runge–Kutta method to solve

$$\frac{dy}{dx} = \frac{y - x}{x + y}, \quad y(0) = 1,$$

at $x = 0.2$, correct to 4 decimal places.

Modified Euler's Method

1. Use Modified Euler's method to compute $y(1.2)$ correct to 5 decimal places, given

$$\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}, \quad y(1) = 1,$$

taking $h = 0.1$.

2. Use Modified Euler's method to compute $y(0.1)$ correct to 4 decimal places, given

$$\frac{dy}{dx} = x - y^2, \quad y(0) = 1,$$

taking $h = 0.1$.

3. Solve using Modified Euler's method

$$y' = 1 + \frac{y}{x}, \quad y(1) = 2,$$

at $x = 1.2$ and $x = 1.4$.

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4. Solve using Modified Euler's method

$$\frac{dy}{dx} = \sin x + \cos y, \quad y(2.5) = 0,$$

at $x = 2.5$.

Milne's Predictor–Corrector Method

1. Given

$$\frac{dy}{dx} = x^2 + \frac{y}{2},$$

and

$$y(1) = 2, \quad y(1.1) = 2.2156, \quad y(1.2) = 2.4649, \quad y(1.3) = 2.7514,$$

compute $y(1.4)$ using Milne's method.

2. Given

$$\frac{dy}{dx} = x - y^2,$$

and

$$y(0) = 0, \quad y(0.2) = 0.02, \quad y(0.4) = 0.0795, \quad y(0.6) = 0.1762,$$

compute $y(0.8)$ using Milne's method.

3. Use Milne's method to find $y(0.8)$, given

$$\frac{dy}{dx} = x + y, \quad y(0) = 1,$$

and

$$y(0.2) = 1.2428, \quad y(0.4) = 1.5836, \quad y(0.6) = 2.0442.$$